Topic: Padding & Strides in CNN

Introduction:

In the previous lecture, you learned that convolution works by sliding a filter (kernel) over an image to produce a **feature map**.

But…

- What happens at the **edges** of the image?
- Why does the **output size shrink** every time?
- How can we **control** how the filter moves?

That's exactly where **padding** and **strides** come in.

Why is Padding Required?

Let's take a simple example:

- Input image: 5×5
- Filter: 3×3
- Stride: 1

Output size = (5 − 3 + 1) = **3×3**

So, we lost the outer pixels of the image because the 3×3 filter cannot fit completely over the edges.

Problem:

Every convolution reduces the image size, which means:

- Information at the borders gets lost.
- After multiple layers, the image shrinks too much.

Solution: Add **padding** — extra pixels around the image (usually zeros) so that the filter can also slide over the edges.

What is Padding?

Padding = adding extra rows and columns (usually zeros) around the image border before convolution.

Let's see the two main types:

1 Valid Padding ("no padding")

- No extra pixels are added.
- Output size **shrinks** after convolution.
- Formula:
- Output size = (Input size - Filter size + 1)

Example:

Input 5×5 → Filter 3×3 → Output 3×3

So, you *lose 1 pixel* from each border.

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2 Same Padding

- Adds enough zeros around the image so that the **output size = input size**.
- Keeps all image features (even borders).
- Commonly used when you want to preserve spatial resolution.

Formula (for stride = 1):

Output size = Input size

Example:

Input 5×5 → Filter 3×3 → Add 1-pixel border → Output 5×5

So, padding helps CNNs "see" **edge pixels** and avoid losing border information.

KERAS Demo

In Keras/TensorFlow:

```
from tensorflow.keras.layers import Conv2D

# Example with padding
Conv2D(filters=32, kernel_size=(3,3), padding='same', strides=1)
```

- padding='same' → keeps the output the same size.
- padding='valid' → no padding (output shrinks).

So, just one line changes how your CNN treats image boundaries.

What are Strides?

Stride controls **how far the filter moves** after each operation.

Let's say stride = 1:

- Filter moves 1 pixel at a time → dense coverage.

If stride = 2:

- Filter skips 1 pixel each time → smaller output.

Example:

- Input 5×5
- Filter 3×3
- Stride 1 → Output 3×3
- Stride 2 → Output 2×2

Increasing stride reduces the **spatial dimension** of the output — like "zooming out".

Special Case with Strides

If the filter size or stride doesn't perfectly fit the input image size,
you may end up with **fractional positions** where the filter doesn't fit completely.

In that case:

- CNNs usually **ignore** the last incomplete step.
- So, some pixels at the end may not be used.

This is why we carefully choose stride and padding to maintain consistent dimensions.

Why are Strides Required?

1. To **control the resolution** of the output feature map.
2. To **reduce computation** (less overlap = faster training).
3. To **summarize features** by skipping redundant areas.

In simple words:

Padding = don't lose border information.
Stride = control how "fast" you move across the image.
Together, they balance **accuracy vs efficiency** in CNNs.

Code Demo

Example:

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D

model = Sequential([
   Conv2D(32, (3,3), strides=(1,1), padding='same', activation='relu',
input_shape=(64,64,3)),
   Conv2D(64, (3,3), strides=(2,2), padding='valid', activation='relu')
])
```

- 1st layer → keeps output same size (due to same padding).
- 2nd layer → halves the output size (due to stride 2 and no padding).

You can visualize this easily by printing model.summary().

Outro (Key Takeaways)

Concept	Meaning	Effect
Padding	Adds extra pixels around image	Preserves border features
Valid Padding	No padding	Output shrinks
Same Padding	Keeps output same size	Retains all info
Stride	Steps taken by filter	Controls output resolution

Large Stride	Faster, less detail	Smaller feature map

Visual Analogy

Imagine scanning a photo with a magnifying glass:

- **Padding:** Adding a frame around the photo so you can scan the very edges too.
- **Stride:** How much you move the magnifying glass each time — one inch or two inches.

Smaller stride → finer detail.

Larger stride → faster but rougher view.